AN EVERYWHERE DEFINED RIGHT DERIVED HOM IS AN EXACT FUNCTOR OF TRIANGULATED CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-existence-of-derived (source lines 5436-5447); the proof environment begins two lines below it and is its own proof, closing at source line 5475.

Editorial note. The proof begins by invoking the author's lemma lemma-construction by label; that lemma's statement and its own proof are retained as a context span placed before the counted statement, so the reference resolves inside the assembled package and no source pointer is declared.

Editorial note. The statement and the proof both display the derived functor notation; the display of the Hom complex into a cone is part of the retained source range and is reproduced as written.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: an everywhere defined right derived Hom is an exact functor of triangulated categories.

Allowed standard prerequisites (editorial).

- Cone and distinguished triangle in a dg module category. The cone of a morphism of dg modules, with its canonical triangle, and the fact that an additive functor compatible with cones respects distinguished triangles.. The identification of the Hom complex into a cone with the cone of the Hom complexes is the exactness argument (source lines 5454-5466).
- Right derived functor and essential constancy. A right derived functor on a triangulated category is computed by an essentially constant diagram indexed by the quasi-isomorphisms out of a representative, when it is everywhere defined.. It is the mechanism that produces the derived functor and the reduction to the abelian-group level (source lines 5467-5474).
- Associated functor on complexes. The author's construction gives a functor from the homotopy category of complexes of A-modules to the homotopy category of dg E-modules (the author's lemma lemma-construction, retained as context).. It is invoked at the start of the proof as the functor whose exactness is established (source line 5452).

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Lemma 0.1. *Let R be a ring. Let $\mathcal{A}$ be a differential graded category over R. Let x be an object of $\mathcal{A}$. Let*

$$(E, d) = \mathrm{Hom}_{\mathcal{A}}(x, x)$$

be the differential graded R-algebra of endomorphisms of x. We obtain a functor

$$\mathcal{A} \longrightarrow Mod^{dg}_{(E,d)}, \quad y \longmapsto \mathrm{Hom}_{\mathcal{A}}(x, y)$$

of differential graded categories by letting E act on $\mathrm{Hom}_{\mathcal{A}}(x, y)$ via composition in $\mathcal{A}$. This functor induces functors

$$Comp(\mathcal{A}) \to Mod_{(A,d)} \quad \text{and} \quad K(\mathcal{A}) \to K(Mod_{(A,d)})$$

by an application of Lemma 0.2.

Proof. This lemma proves itself. □

Lemma 0.2. *Let R be a ring. A functor $F : \mathcal{A} \to \mathcal{B}$ of differential graded categories over R induces functors $Comp(\mathcal{A}) \to Comp(\mathcal{B})$ and $K(\mathcal{A}) \to K(\mathcal{B})$.*

Proof. Omitted. □

Lemma 0.3. *In the situation above. If the right derived functor $R\mathrm{Hom}(K^\bullet, -)$ of $\mathrm{Hom}(K^\bullet, -) : K(\mathcal{A}) \to D(Ab)$ is everywhere defined on $D(\mathcal{A})$, then we obtain a canonical exact functor*

$$R\mathrm{Hom}(K^\bullet, -) : D(\mathcal{A}) \longrightarrow D(E, d)$$

of triangulated categories which reduces to the usual one on taking associated complexes of abelian groups.

Proof. Note that we have an associated functor $K(\mathcal{A}) \to K(\mathrm{Mod}_{(E,\mathrm{d})})$ by Lemma 0.1. We claim this functor is an exact functor of triangulated categories. Namely, let $f : A^\bullet \to B^\bullet$ be a map of complexes of $\mathcal{A}$. Then a computation shows that

$$\mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{A})}(K^\bullet, C(f)^\bullet) = C\left(\mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{A})}(K^\bullet, A^\bullet) \to \mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{A})}(K^\bullet, B^\bullet)\right)$$

where the right hand side is the cone in $\mathrm{Mod}_{(E,\mathrm{d})}$ defined earlier in this chapter. This shows that our functor is compatible with cones, hence with distinguished triangles. Let $X^\bullet$ be an object of $K(\mathcal{A})$. Consider the category of quasi-isomorphisms $s : X^\bullet \to Y^\bullet$. We are given that the functor $(s : X^\bullet \to Y^\bullet) \mapsto \mathrm{Hom}_{\mathcal{A}}(K^\bullet, Y^\bullet)$ is essentially constant when viewed in $D(Ab)$. But since the forgetful functor $D(E, \mathrm{d}) \to D(Ab)$ is compatible with taking cohomology, the same thing is true in $D(E, \mathrm{d})$. This proves the lemma. □